

RealEstate Platform: Windows Setup Guide

Step-by-step instructions for a developer who downloads the project and wants to run the backend, frontend, MySQL database, optional MongoDB, and Postman APIs in VS Code.

1. Install Required Apps

Install these applications first:

- Node.js LTS: <https://nodejs.org/>
- MySQL Community Server: <https://dev.mysql.com/downloads/mysql/>
- MySQL Workbench: <https://dev.mysql.com/downloads/workbench/>
- VS Code: <https://code.visualstudio.com/>
- Optional MongoDB Community Server: <https://www.mongodb.com/try/download/community>

During MySQL installation, keep port 3306, set a root password, and remember that password.

2. Open Project In VS Code

Open VS Code, then use File > Open Folder and select the lab-2 project folder. Open a terminal with Terminal > New Terminal.

3. Check MySQL Works

In the VS Code terminal, run:

```
mysql -u root -p
```

Enter your MySQL root password. If you see mysql>, MySQL is working. Exit with:

```
exit;
```

If Windows says mysql is not recognized, use MySQL Workbench or add MySQL to PATH. Common path:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin
```

4. Create Backend Environment File

From the project root in VS Code terminal:

```
cd backend  
copy .env.example .env
```

Open backend\.env and configure:

```
PORT=5001  
FRONTEND_URL=http://localhost:3000  
  
DB_HOST=localhost  
DB_USER=root  
DB_PASSWORD=your_mysql_password
```

```
DB_NAME=realestate_db

JWT_SECRET=change_this_to_any_long_secret
JWT_EXPIRES_IN=7d
NODE_ENV=development
```

5. Create Frontend Environment File

```
cd ../frontend
copy .env.example .env
```

Open frontend\.env and keep:

```
REACT_APP_API_BASE=http://localhost:5001/api
```

6. Create Database And Seed Data

Go back to the project root:

```
cd ..
```

Run the full database setup:

```
mysql -u root -p < database\run-all.sql
```

Enter your MySQL password. This creates realestate_db, tables, migrations, demo users, properties, agencies, agents, and plans.

If mysql is not recognized, run with the full path:

```
& "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -p <
database\run-all.sql
```

7. Install Backend Dependencies

```
cd backend
npm install
```

8. Install Frontend Dependencies

```
cd ../frontend
npm install
```

9. Run Backend

Open VS Code terminal 1:

```
cd backend
npm run dev
```

You should see:

```
Database connected successfully to "realestate_db".
Server running on http://localhost:5001
```

Test in browser:

```
http://localhost:5001/api
```

10. Run Frontend

Open VS Code terminal 2:

```
cd frontend
npm start
```

Open:

```
http://localhost:3000
```

11. Login Demo Accounts

Use one of these accounts:

Role	Email	Password
Admin	admin@realestate.local	Admin123!
Seller	seller@realestate.local	Seller123!
Buyer	buyer@realestate.local	Buyer123!

12. Optional MongoDB On Windows

MongoDB is optional. Only install it if you want Mongo view logs, archives, and Mongo-backed top property reports.

During MongoDB installation, choose Complete setup, install MongoDB as a Windows Service, and keep the default port 27017.

Check MongoDB:

```
mongosh
use realestate
show collections
exit
```

To enable MongoDB, add this to backend\.env:

```
MONGO_URI=mongodb://localhost:27017/realestate
MONGO_ARCHIVE_ENABLED=true
MONGO_ARCHIVE_CRON=0 3 * * *
```

If not using MongoDB, keep:

```
MONGO_URI=
MONGO_ARCHIVE_ENABLED=false
```

Restart the backend after changing .env.

13. VS Code Database Extensions

For MySQL, install these VS Code extensions:

- SQLTools
- SQLTools MySQL/MariaDB Driver

Create a connection:

```
Connection type: MySQL/MariaDB
Name: RealEstate Local
```

```
Server: localhost
Port: 3306
Database: realestate_db
Username: root
Password: your_mysql_password
```

Optional Mongo extension:

```
MongoDB for VS Code
Connection string: mongodb://localhost:27017
```

14. Postman API Testing

Import this file into Postman:

```
docs\postman-collection.json
```

Set collection variables:

```
baseUrl = http://localhost:5001/api
token = token_from_login
```

Get token using:

```
POST {{baseUrl}}/auth/login

{
  "email": "admin@realestate.local",
  "password": "Admin123!"
}
```

15. Common Problems

- Backend MySQL connection failed: check MySQL is running, DB_PASSWORD is correct, and database/run-all.sql was executed.
- Frontend failed to fetch: check backend is running on port 5001 and frontend/.env has REACT_APP_API_BASE=http://localhost:5001/api.
- Protected API calls fail: login again and update the Postman token variable.
- MongoDB fails: it is optional. Set MONGO_URI empty and MONGO_ARCHIVE_ENABLED=false, then restart backend.